

Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization

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Abstract—State-of-the-art optical wireless positioning (OWP) commonly reaches centimeter-level accuracy by depending on dense multi-light-emitting diodes (LED) infrastructures, photodiode (PD) arrays, or image-sensor receivers, incurring hardware complexity and deployment cost. This paper introduces a single beam-steered LED, single-PD OWP architecture that achieves three-dimensional (3D) localization without receiver rotation, cameras, or PD arrays; the core idea is to steer the transmitter through K known orientations and exploit the resulting received-signal-strength variations at the PD to estimate LED-to-PD direction and distance. We derive a composite Cramér–Rao lower bound and position-error bound (PEB) for the joint observation model, and cast the steering-pattern design as a genetic algorithm that minimizes the PEB over a 3D testbed. We develop both model-based a constrained nonlinear estimator and closed-form direction estimators: a statistically efficient generalized least squares solution, and a lightweight weighted least squares approximation. Simulations demonstrate centimeter-level accuracy for 3D OWP with a single beam-steered LED and a single PD.

Index Terms—3D indoor localization, beam steering, Cramér–Rao bounds, optical wireless communications, optical wireless positioning

I. INTRODUCTION

GLOBAL navigation satellite systems (GNSS) provide ubiquitous outdoor positioning but are unreliable indoors due to severe attenuation, blockage, and multipath. Therefore, there is a need for dedicated indoor position systems (IPS), which use radio-frequency techniques leveraging existing communications infrastructure, such as Wi-Fi [1], Bluetooth Low Energy (BLE) beacons [2], ultra-wideband (UWB) [3], radio-frequency identification (RFID) [4], and emerging 5G positioning [5], are widely studied and deployed [5], [6]. However, in cluttered interiors they are strongly affected by multipath, non-line-of-sight (NLOS) conditions, and time-varying interference in shared spectrum, typically yielding meter-level accuracy unless dense infrastructure, calibration, or specialized hardware is used [7], [8].

In this context, optical wireless positioning (OWP) leverages low-cost, small-form-factor light-emitting diodes (LEDs) that transmit modulated optical signals, often imperceptible to humans, captured by photodiodes (PDs) or cameras to transform lighting into a positioning network [9]. The spatial confinement of light (without wall penetration), immunity to RF electromagnetic interference, high spatial reuse, and

more deterministic line-of-sight (LOS) optical channels enable centimeter-level localization with cost-effective luminaires [10]. This makes OWP attractive for navigation and asset tracking in malls, airports, and museums where GNSS is unavailable, as well as in electromagnetic-sensitive environments such as hospitals and airplane cabins [9]. Moreover, the same LED infrastructure can provide optical wireless communication (OWC), often termed light-fidelity (LiFi) [11], so that illumination, data, and positioning coexist in a unified platform. In this work, we deliberately employ a near-infrared (NIR) LED transmitter for positioning, as opposed to conventional visible-light positioning (VLP), which must respect illumination standards and uniformity when reusing luminaires. Since NIR is imperceptible to humans, the positioning link is decoupled from visible-light illumination constraints. This allows us to steer and reorient the source without affecting room lighting [12].

OWP follows two paradigms: data-driven and model-based. Data-driven methods (such as fingerprinting with k -nearest neighbors, support vector machines, random forests, gradient boosting, Gaussian process regression, multilayer perceptrons, convolutional neural networks for structured PD signals, recurrent/temporal models, and transformer architectures [13], [14]) are effective with abundant labeled data, but these black-box approaches are often opaque, sensitive to domain shift (hardware, illumination, occlusion), and require periodic retraining, offering limited physical guarantees [15]. By contrast, model-based estimators grounded in radiometry and geometry (e.g., Lambertian propagation, field-of-view constraints) provide interpretability, sample efficiency, analyzable uncertainty, and reliable extrapolation, and they support joint system co-design [16].

Most existing OWP implementations rely on traditional multi-LED transmitter constellations, where multiple fixed LEDs with known positions act as anchors [29]. Using multiple LEDs allows the receiver to compute its position via triangulation, trilateration, or multilateration, analogous to how GNSS uses many satellites [9]. OWP implementations, specifically VLP, reuses luminaires, and the LED constellation is co-designed with illumination. This co-design includes placement, power, and beam patterns that must satisfy lighting standards and uniformity, which constrains anchor geometry and modulation [30]. However, these multi-LED requirements complicate practical deployment. They presume a dense infrastructure; at least four LEDs must be concurrently visible to the receiver to resolve a three dimensional (3D) position [31]–[34], which may not be true in environments with only sparse lighting. Installing and maintaining a dense LED network, with unique IDs or modulation for each lamp,

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TABLE I
STATE-OF-THE-ART SINGLE-LED OWP SYSTEMS AND REPORTED PERFORMANCE.

Year	Work	Configuration	Rx Orientation	2D/3D	Method	Error (APE)	Room (m^3)	Notes
2020	[17]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Model-based (Geometrical equations)	2.52 cm	$1 \times 1 \times 1.5$	Sim.
2021	[18]	1 LED / 3 tilted PD	Fixed	2D	Long Short Term Memory- Fully Connected Network	0.92 cm	$1 \times 1.1 \times 1.75$	Exp.
2021	[19]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	2D	Bayesian algorithm	9.27 cm	$3 \times 3 \times 3$	Exp.
2022	[20]	1 LED / 1 rotatable PD	Fixed	2D	Extreme Learning Machine	1.74 cm	$4 \times 4 \times 3.1$	Sim.
2023	[21]	1 LED / 4 PD	Fixed	2D	Least Square	7 cm (CDF _{90%})	$6 \times 6 \times 3$	Exp.
2024	[22]	1 reorientable LED / 1 PD	Arbitrary	2D	Model-based (Power ratio)	5.9 cm (CDF _{90%})	$4 \times 4 \times 2.5$	Sim.
2024	[23]	1 LED / 4 PD	Fixed	2D	Extreme Learning Machine / WKNN	2.93 cm	$5 \times 5 \times 3$	Sim.
2024	[24]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Deep Residual Shrinkage method	5.85 cm	$3.6 \times 3.6 \times 3.0$	Exp.
2024	[25]	1 LED / 1 PD (tilted-rotatable PD)	Rotatable	2D	Least Squares + Improved Grey Wolf Optimizer (LS-IGWO)	1.65 cm	$0.5 \times 0.5 \times 0.3$	Exp.
2024	[26]	1 LED / 4 PD + IMU (orientation)	Arbitrary	3D	MultiLayer Perceptron	1.77 cm	$1.5 \times 1.5 \times 1.5$	Exp.
2025	[27]	1 LED / 1 rotatable PD	Fixed	2D	Least Square	4.96 cm	$5 \times 5 \times 4$	Exp.
2025	[28]	1 LED / 16 IRS / 1 PD	Fixed	2D	Levenberg–Marquardt algorithm	5.6 cm (RMSE)	$4 \times 5 \times 5$	Sim.
2025	Ours	1 reorientable LED / 1 PD	Arbitrary	3D	Generalised Least Squares	1.54 cm	$3 \times 3 \times 2$	Sim.

Note: “Sim.” denotes simulation results; “Exp.” denotes experimental results.

increases installation cost and complexity [35]. Consequently, reducing the infrastructure burden by localizing with single light sources is a highly desirable goal in the field [13].

Recent research has explored OWP schemes that break from the multi-LED paradigm. One notable direction is single-LED positioning, which aims to achieve localization with only one transmitter [17]–[28]. Table I summarizes representative single-LED OWP systems, reports the average positioning error (APE), and highlights the diversity of receiver designs and estimation methods: many works employ multiple PDs (e.g., 4 PDs or 3 tilted PDs) [17], [19], [21], [23], [24], others rely on a single rotatable PD [20], [25], [27], and some integrate aiding sensors such as an inertial measurement unit [26] or virtualize anchors via intelligent reflecting surfaces (IRS) [28]. Regarding dimensionality, several demonstrations are two dimensional (2D) [18]–[23], [25], [27], [28], whereas fewer achieve 3D with a single LED [17], [24], [26]. A single light source can simplify infrastructure and calibration, but at a fixed orientation one measurement is insufficient to resolve a 3D position. Therefore, single-LED systems typically extract additional information via receiver diversity/rotation, sensor fusion, or engineered propagation (IRS) on the UE side [17], [18], [20], [25], [26], [28]. These limitations motivate the search for a new 3D single-LED single-PD positioning framework.

In parallel, recent advances in OWC have delivered agile beam-steering transmitters. For example, this includes high-speed microelectromechanical systems micromirror gimbals [36], liquid-crystal or metasurface deflectors that widen the field-of-view without sacrificing speed [37], silicon-photonics optical-phased arrays that shift their beam tens of degrees in micro-seconds with no moving parts [38]. In addition, recent OWC/LiFi roadmaps increasingly regard dynamic beam steering as a core enabler for high-capacity, mobile, multiuser links and blockage mitigation making it natural to exploit the same capability for positioning [38]–[40]. These demonstrations confirm that equipping a single indoor luminaire with a steering mechanism is feasible and highly synergistic.

In this work, we study a novel model-based beam-steered single-LED single-PD OWP architecture for 3D indoor localization. The main contributions are summarized as follows:

- A novel single-LED single-PD 3D OWP framework that achieves full 3D localization without receiver rotation, camera sensing, or PD arrays, enabled by steering the optical beam of a single LED in K different directions.
- Cramér–Rao lower bound (CRLB) / position error bound (PEB) analysis for a composite observation model. We derive Fisher-information expressions, and we characterize identifiability and Fisher information matrix (FIM) rank conditions as a function of K .
- Genetic algorithm (GA)-based orientation-set design. We cast the transmitter orientation selection as a GA optimization that minimizes the root-mean-square error (RMSE) of PEB over a 3D testbed, yielding practical design rules for the choice of K and set of orientations.
- Closed-form linear direction estimators. Using a ratio-based linearization, we develop a statistically efficient generalized least squares (GLS) estimator, and a lightweight weighted least squares (WLS) approximation, and we evaluate their computational latency.

This paper is organized as follows: Section II presents the system model and the proposed 3D localization procedure. Section III derives the PEB for single-LED positioning. Section IV optimizes the transmitter beam-orientation set. Section V develops a nonlinear (NL) direction estimator. Section VI introduces closed-form linear estimators (GLS and WLS). Section VII reports estimators comparison. Section VIII concludes the paper.

Notation: Bold lowercase letters denote column vectors (e.g., $\mathbf{x}$) and bold uppercase letters denote matrices (e.g., $\mathbf{A}$); scalars are in lightface. The Euclidean norm is $\|\cdot\|$, the inner product is $\mathbf{a} \cdot \mathbf{b}$, the trace is $\text{tr}(\cdot)$, and the gradient with respect to $\mathbf{r} = [x, y, z]$ is $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^T$. The transpose is $(\cdot)^T$, the identity matrix of dimension n and the $n \times m$ zero matrix are $\mathbf{I}_n$ and $\mathbf{0}_{n \times m}$, respectively, and $\text{diag}(\cdot)$ forms a diagonal matrix from its arguments. The smallest

eigenvalue and a corresponding unit-norm eigenvector of a symmetric matrix $\mathbf{M}$ are denoted by $\lambda_{\min}(\mathbf{M})$ and $\mathbf{v}_{\min}(\mathbf{M})$; the Loewner order $\succeq$ indicates positive semidefiniteness. $\mathbb{R}^n$ is the n -dimensional real space and $\mathbb{S}^2$ the unit sphere in $\mathbb{R}^3$. Random variables follow the convention $x \sim \mathcal{N}(\mu, \sigma^2)$, with expectation $\mathbb{E}[\cdot]$ and variance $\text{Var}[\cdot]$. A hat $\hat{(\cdot)}$ denotes an estimate and an overbar $\bar{(\cdot)}$ a sample mean.

II. SYSTEM MODEL AND PROPOSED 3D LOCALIZATION METHOD

A. System Model

Figure 1 illustrates the single-LED single-PD OWP setup analyzed in this work. A LED transmitter is mounted on the ceiling at a known position $\mathbf{t} = [0, 0, H]^\top$ and can steer its optical axis along a unit vector $\mathbf{n}_{t,i} \in \mathbb{S}^2$ for the i -th orientation, $i = 1, \dots, K$. The receiver is a single PD with fixed normal along the vertical axis $\mathbf{n}_r = [0, 0, 1]^\top$ located at an unknown position $\mathbf{r} = [x, y, z]^\top$. Let $\mathbf{d} = \mathbf{r} - \mathbf{t}$ denote the transmitter–receiver displacement vector, $d = \|\mathbf{d}\|$ the separation distance, and $\mathbf{n}_d = \mathbf{d}/d$ the corresponding unit direction vector.

For the i -th LED orientation, the received signal strength (RSS) at the PD is modeled as [41]:

$$S_{r,i} = R_p P_t h_{\text{LOS},i} + w_i, \quad (1)$$

where R_p is the PD responsivity, P_t is the transmitted radiant power, and $w_i \sim \mathcal{N}(0, \sigma_w^2)$ models a zero-mean additive white Gaussian noise (AWGN) term arising primarily from shot and Johnson noise, and assumed independent and identically distributed (i.i.d.) across orientations and time (homoscedastic), and $h_{\text{LOS},i}$ denotes the line-of-sight (LOS) channel gain, which for a Lambertian source is given by [41]:

$$h_{\text{LOS},i} = \begin{cases} \frac{(m+1)A_{\text{det}}}{2\pi d^2} \cos^m(\phi_i) \cos(\psi), & \psi \leq \Psi_{\text{FOV}}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where m is the Lambertian order of the LED, related to the semi-angle at half power $\Phi_{1/2}$ by $m = -\ln 2 / \ln(\cos(\Phi_{1/2}))$, A_{det} the PD effective area, Ψ_{FOV} the receiver field of view, and the irradiance/incidence cosines geometrically given by:

$$\cos \phi_i = \frac{\mathbf{n}_{t,i} \cdot \mathbf{d}}{d}, \quad \cos \psi = \frac{\mathbf{n}_r \cdot (-\mathbf{d})}{d}. \quad (3)$$

Dividing (1) by R_p yields the received optical power:

$$P_{r,i} = P_t h_{\text{LOS},i} + n_i, \quad (4)$$

where $n_i \sim \mathcal{N}(0, \sigma^2)$ is zero-mean AWGN, remains i.i.d. and homoscedastic, with variance $\sigma^2 = (\sigma_w^2 / R_p^2)$.

At every transmitter orientation i , the receiver acquires N independent samples. The sample mean of the received power is:

$$\bar{P}_{r,i} = \frac{1}{N} \sum_{k=1}^N P_{r,i,k}, \quad (5)$$

where $P_{r,i,k}$ denotes the k -th instantaneous received power sample at orientation i , corresponding to the k -th realization of the random variable $P_{r,i}$ in (4). Under i.i.d. Gaussian noise, $\bar{P}_{r,i}$ is the minimum-variance unbiased (MVU) estimator of the received power.

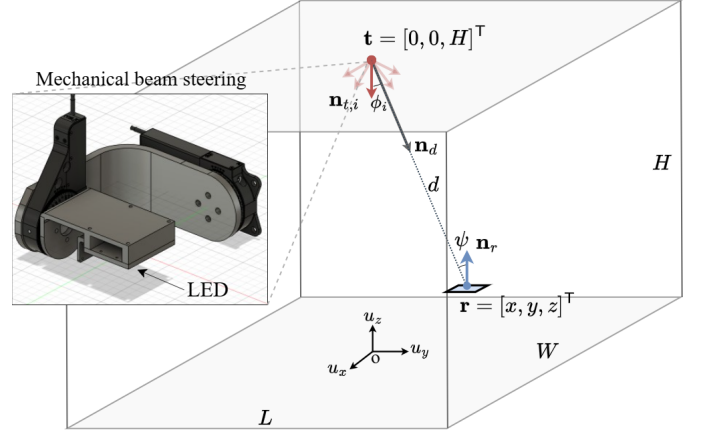


Fig. 1. Single beam-steered LED / single-PD OWP geometry. A beam-steered LED at $\mathbf{t} = [0, 0, H]^\top$ (e.g., mechanical steering) points along $\{\mathbf{n}_{t,i}\}_{i=1}^K$; the PD is at $\mathbf{r} = [x, y, z]^\top$.

The corresponding signal-to-noise ratio (SNR), given the receiver position and a specific transmitter orientation i , noted $\text{SNR}_{r,i}$, as well as the average SNR for a given receiver position, noted SNR_r , and for the entire testbed, noted SNR , are respectively defined as:

$$\text{SNR}_{r,i} = \frac{(R_p P_t h_{\text{LOS},i})^2}{\sigma_w^2}, \quad (6a)$$

$$\text{SNR}_r = \frac{1}{K} \sum_{i=1}^K \text{SNR}_{r,i}, \quad (6b)$$

$$\text{SNR} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{SNR}_r, \quad (6c)$$

where $\mathcal{R}$ denotes the set of receiver testbed positions and $|\mathcal{R}|$ their cardinality.

B. General 3D Single-LED Positioning Procedure

The proposed localization process comprises two consecutive stages:

1) *Direction Finding*: The first stage estimates the direction vector $\mathbf{n}_d$ by steering the LED through the K predefined orientations $\{\mathbf{n}_{t,i}\}_{i=1}^K$ and processing the power means $\{\bar{P}_{r,i}\}_{i=1}^K$. Section V (NL estimator) and Section VI (linear GLS/WLS estimators) detail statistically grounded solutions.

2) *Distance Recovery*: After direction finding, the LED is aligned with the estimated direction $\hat{\mathbf{n}}_d$ and, for beam-forming symmetry, the receiver is oriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$. Under this beam-steered configuration and using (2), (3), and (4), the received power for this new orientation ($K+1$), denoted as $P_{r,K+1}$, obeys:

$$P_{r,K+1} = \frac{C}{d^2} + n_{K+1}, \quad (7)$$

where C is an optical constant calibrated once per hardware set-up:

$$C = \frac{P_t(m+1)A_{\text{det}}}{2\pi}, \quad (8)$$

and n_{K+1} is the noise term for orientation ($K+1$), following the same statistical properties as the noise described in (4).

Collecting N independent samples $\{P_{r,K+1,j}\}_{j=1}^N$ and using (5), the sample mean satisfies $\bar{P}_{r,K+1} \sim \mathcal{N}(C/d^2, \sigma^2/N)$. Substituting $\bar{P}_{r,K+1}$ into (7) provides the distance estimate:

$$\hat{d} = \sqrt{\frac{C}{\bar{P}_{r,K+1}}}, \quad (9)$$

and, finally, the 3D estimated position of the receiver, denoted as $\hat{\mathbf{r}}$, is $\hat{\mathbf{r}} = \mathbf{t} + \hat{d}\hat{\mathbf{n}}_d$.

III. POSITION ERROR BOUND

This section derives the PEB, i.e., the CRLB specialised to the receiver's 3D coordinate vector $\mathbf{r} = [x, y, z]^\top$. The PEB quantifies the minimum achievable variance of any unbiased estimator of $\mathbf{r}$ under the proposed OWP model and serves both as a benchmark for estimator performance and as an objective for system design.

A. Joint Statistical Model of the Observations

According to Section II, the sample-mean optical power, as defined in (5), for each of the K transmitter orientations used during the direction finding stage is distributed as:

$$\bar{P}_{r,i} \sim \mathcal{N}(\mu_i(\mathbf{r}), \sigma^2/N), \quad i = 1, \dots, K, \quad (10)$$

where $\mu_i(\mathbf{r}) = P_t h_{\text{LOS},i}(\mathbf{r})$ and $h_{\text{LOS},i}$ is defined in (2) with the cosines in (3). During the distance recovery stage, the mean optical power in aligned configuration obeys:

$$\bar{P}_{r,K+1} \sim \mathcal{N}(\mu_{K+1}(\mathbf{r}), \sigma^2/N), \quad (11)$$

where $\mu_{K+1}(\mathbf{r}) = C/d^2(\mathbf{r})$, $d(\mathbf{r}) = \|\mathbf{r} - \mathbf{t}\|$ and C is the optical constant introduced in (8).

Stacking these $K+1$ independent observations into $\mathbf{z} = [\bar{P}_{r,1}, \dots, \bar{P}_{r,K}, \bar{P}_{r,K+1}]^\top$ yields the multivariate Gaussian model:

$$\mathbf{z} = \{z_i\}_{i=1, \dots, K+1} \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{r}), (\sigma^2/N) \mathbf{I}_{K+1}), \quad (12)$$

with mean vector $\boldsymbol{\mu}(\mathbf{r}) = [\mu_1(\mathbf{r}), \dots, \mu_{K+1}(\mathbf{r})]^\top$ and diagonal covariance σ^2/N .

B. Log-Likelihood and Score Function

Under the assumption of independent Gaussian measurements, the log-likelihood of $\mathbf{r}$ given $\mathbf{z}$ is:

$$\ell(\mathbf{r}) = \ln L(\mathbf{r} | \mathbf{z}) = -\frac{N}{2\sigma^2} \sum_{i=1}^{K+1} [z_i - \mu_i(\mathbf{r})]^2 + c_0, \quad (13)$$

where c_0 is an additive constant independent of $\mathbf{r}$ [42]. Differentiating $\ell(\mathbf{r})$ with respect to $\mathbf{r}$ defines the score vector:

$$\mathbf{s}(\mathbf{r}) = \nabla_{\mathbf{r}} \ell(\mathbf{r}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} [z_i - \mu_i(\mathbf{r})] \nabla_{\mathbf{r}} \mu_i(\mathbf{r}), \quad (14)$$

where $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^\top$ is the gradient operator.

C. Fisher Information Matrix

For distributions in the exponential family, such as the homoscedastic Gaussian model assumed here, the FIM can be obtained as the variance of the score vector, avoiding second-order differentiation and yielding greater numerical stability [43]:

$$\mathcal{I}(\mathbf{r}) = \text{Var}[\mathbf{s}(\mathbf{r})] = \mathbb{E}[\mathbf{s}(\mathbf{r}) \mathbf{s}(\mathbf{r})^\top] \quad (15)$$

$$= \frac{N}{\sigma^2} \sum_{i=1}^{K+1} [\nabla_{\mathbf{r}} \mu_i(\mathbf{r})] [\nabla_{\mathbf{r}} \mu_i(\mathbf{r})]^\top. \quad (16)$$

The second equality follows from $\mathbb{E}[z_i] = \mu_i(\mathbf{r})$, which implies $\mathbb{E}[(z_i - \mu_i)(z_j - \mu_j)] = 0$ for $i \neq j$. Hence the total information is the coherent superposition of the differential contributions from each LED orientation and the range measurement. Required gradients are:

1) *Direction-Finding Measurements*: For the i -th orientation:

$$\nabla_{\mathbf{r}} \mu_i(\mathbf{r}) = P_t \nabla_{\mathbf{r}} h_{\text{LOS},i}(\mathbf{r}), \quad (17)$$

which, by differentiating (2) via the chain rule on $d(\mathbf{r})$, $\cos \phi_i$ and $\cos \psi$, yields the closed-form:

$$\nabla_{\mathbf{r}} \mu_i(\mathbf{r}) = \frac{C}{d^3} \left[m \cos^{m-1} \phi_i \cos \psi \mathbf{n}_{t,i} - \cos^m \phi_i \mathbf{n}_r - (m+3) \cos^m \phi_i \cos \psi \mathbf{n}_d \right], \quad (18)$$

which vanishes whenever $\psi > \Psi_{\text{FOV}}$ since $h_{\text{LOS},i} = 0$ outside the receiver's field of view.

2) *Distance-Recovery Measurement*:

$$\nabla_{\mathbf{r}} \mu_{K+1}(\mathbf{r}) = -\frac{2C}{d^4} (\mathbf{r} - \mathbf{t}) = -\frac{2C}{d^3} \mathbf{n}_d, \quad (19)$$

where $\mathbf{n}_d$ is the unit direction vector defined in Section II.

D. Formal Definition of the PEB

Let $\hat{\mathbf{r}}$ be any unbiased estimator of $\mathbf{r}$. The Cramér-Rao inequality asserts: $\text{Cov}[\hat{\mathbf{r}}] \succeq \mathcal{I}^{-1}(\mathbf{r})$. Accordingly, a scalar performance metric, the PEB, is defined as:

$$\text{PEB}(\mathbf{r}) = \sqrt{\text{tr}\{\mathcal{I}^{-1}(\mathbf{r})\}}. \quad (20)$$

The square root of the trace of $\mathcal{I}^{-1}$ corresponds to the minimum achievable RMSE in position estimation for the described OWP configuration. A small PEB indicates that the chosen set of LED orientations and the range measurement provide sufficient and complementary information for high-precision localization; conversely, a large PEB suggests the need to increase K , improve the SNR, or redesign the orientation pattern.

E. Number of LED Orientations

1) *Underdetermined Regime* ($K \leq 2$): In this regime, the Fisher information matrix $\mathcal{I}(\mathbf{r})$ from (16) is rank-deficient. Each of the K orientation measurements contributes at most to one independent gradient vector, and the distance-recovery measurement adds a single additional vector. When $K \leq 2$, there are fewer than three linearly independent gradients in total, so $\mathcal{I}(\mathbf{r})$ is singular and the CRLB on any unbiased position estimate diverges [29].

2) *Just-Determined Regime* ($K = 3$): With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the 3D parameter space. In generic geometries the FIM is full rank and $\mathcal{I}^{-1}(\mathbf{r})$ exists, yielding a finite PEB. However, the absence of redundancy makes the bound highly sensitive to small perturbations in orientation geometry [18].

3) *Overdetermined Regime* ($K \geq 4$): For four or more orientations, the FIM accumulates redundant outer-products of gradients. This overdetermination improves the numerical conditioning of $\mathcal{I}(\mathbf{r})$ and generally lowers the PEB, since extra measurements provide additional independent information beyond the minimal requirement [9], [26].

IV. OPTIMIZATION OF THE ORIENTATION SET

Equation (18) shows that the FIM $\mathcal{I}(\mathbf{r})$ depends explicitly on the transmitter orientation vectors $\{\mathbf{n}_{t,i}\}_{i=1}^K$, and hence so does the PEB in (20). We thus define the objective function as the RMSE of the PEB values evaluated over all receiver positions within the 3D testbed specified in Table II. Selecting the orientation set that minimizes this objective is essential for localization accuracy. However, $\text{PEB}(\{\mathbf{n}_{t,i}\})$ is nonlinear in the orientation parameters, involves multiple coupled terms (e.g., powers of cosines and distance-dependent factors), and exhibits nonsmooth behavior due to the receiver FOV. Consequently, closed-form optimization is not available and purely local methods are prone to suboptimal solutions. To robustly explore the parameter space and approach a globally optimal orientation set for a given K , we employ a GA [44].

TABLE II
PARAMETERS USED FOR GA OPTIMIZATION.

System Parameters	
Parameter	Value
Transmitter coordinates ($\mathbf{t}$)	$[0, 0, 2]^T$ m
Transmitted optical power (P_t)	0.405 W
Half-power semi-angle ($\Phi_{1/2}$)	$\{30^\circ, 45^\circ, 60^\circ\}$
Lambertian order (m)	$-\ln 2 / \ln(\cos(\Phi_{1/2}))$
Photodiode effective area (A_{det})	4.8×5.5 mm ²
Receiver field of view (Ψ_{FOV})	85°
Electrical noise variance (σ_w^2)	1.19×10^{-14} A ²
PD responsivity (R_p)	0.63 A/W
Noise variance (σ^2)	3×10^{-14} W ²
Samples per orientation (N)	1000
Genetic Algorithm	
Parameter	Value
Population size (P)	300
Maximum generations (G_{max})	200
Crossover probability (p_c)	0.8
Search range (Ω)	$\theta_i \in [0, 80^\circ], \varphi_i \in [0, 360^\circ]$
Objective function ($f(\mathbf{x})$)	RMSE of the PEB
Decision variables ($\mathbf{x}$)	$[\theta_1, \varphi_1, \dots, \theta_K, \varphi_K]^T$
3D Testbed	
Parameter	Value
Room dimensions ($L \times W \times H$)	$3 \times 3 \times 2$ m ³
Range in x, y	$[-1.5, 1.5]$ m
Range in z	$[0, 1.2]$ m
Grid step	0.2 m
Total number of points tested	1792

Within this GA framework, each individual encodes K steering directions via tilt-azimuth pairs $\{(\theta_i, \varphi_i)\}_{i=1}^K$, yielding a $2K$ -dimensional genotype $\mathbf{x} = [\theta_1, \varphi_1, \dots, \theta_K, \varphi_K]^T$.

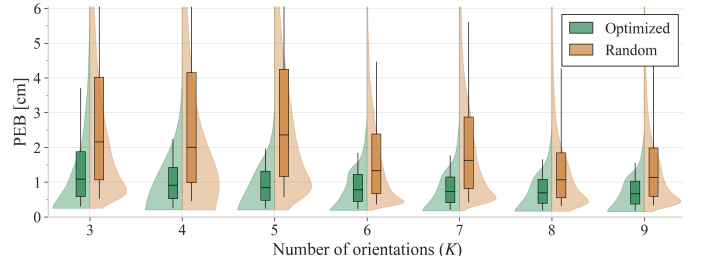


Fig. 2. Distribution of the PEB across all receiver locations in the $3 \times 3 \times 2$ m³ testbed. For each K , green shows the GA-optimized orientation set and orange a randomly selected set. Boxes and whiskers summarize the sample distribution, while the shaded profiles depict the corresponding PDF.

We adopt a nadir-referenced spherical parameterisation where θ_i is the tilt from $-\mathbf{u}_z$ and φ_i is the azimuth in the $\mathbf{u}_x$ – $\mathbf{u}_y$ plane measured counterclockwise from $+\mathbf{u}_x$, with domains $\theta_i \in [0, \theta_{\text{max}}]$ and $\varphi_i \in [0^\circ, 360^\circ]$. The mapping to unit-norm orientation vectors in the reference frame $(\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z)$ is:

$$\mathbf{n}_{t,i} = [\sin \theta_i \cos \varphi_i, \sin \theta_i \sin \varphi_i, -\cos \theta_i]^T. \quad (21)$$

Spherical coordinates are preferred to Cartesian because they describe unit-norm directions with two angles, avoiding the nonlinear unit-norm constraint and providing bounded domains that simplify GA search. Standard genetic operators—tournament selection, crossover with probability p_c , and adaptive feasible mutation—are applied for at most G_{max} generations, with early termination when the average relative improvement of the best fitness over G_{max} generations falls below 10^{-6} . The implementation is in MATLAB; optimization parameters are summarized in Table II.

Table III lists the optimal K -orientation sets for the system parameters in Table II. Each set is unique and no set is a subset of another. In Fig. 2, for each K we present box-and-whisker plots of the PEB computed over all receiver locations in the $3 \times 3 \times 2$ m³ testbed; optimized sets are shown in green, randomly selected sets in orange, and the shaded profiles depict the corresponding probability density function (PDF) estimates. Quantitatively, the optimized sets achieve a median of PEB below 1.5 cm for all $K \in \{3, \dots, 9\}$ (worst median 1.43 cm at $K = 3$) and reach 0.74 cm at $K = 9$. The GA consistently outperforms random orientations: across K , the optimized sets reduce the median PEB by 49%–72% and exhibit smaller interquartile ranges. These gains are reflected in the distributions, with the central tendency improving monotonically as K increases.

Figure 3 presents spatial heat maps of the PEB across a 3×3 m² floor at $z = 0.8$ m for $K = 5$ orientations with the transmitter at the room centre. In the optimal configuration, shown in Fig. 3(a), the highest errors occur uniformly along the boundaries and especially at the corners, analogous to multi-transmitter layouts [45], but remain spatially consistent. By contrast, the random configuration in Fig. 3(b) exhibits both larger and more irregular error patterns.

We now turn from optimization to performance characterization: using the optimized orientation sets, we quantify how the 90th-percentile of the PEB distribution ($\text{PEB}_{90\%}$) varies with K , $\Phi_{1/2}$, and SNR. Figure 4 depicts the $\text{PEB}_{90\%}$

TABLE III
OPTIMAL ORIENTATIONS (θ_i, φ_i) IN DEGREES $[\circ]$ FOR VARIOUS K IN A $3 \times 3 \times 2 \text{ m}^3$ ROOM.

K	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$	$i = 8$	$i = 9$
3	(35.4, 140.1)	(33.3, 36.4)	(29.6, 262.7)	-	-	-	-	-	-
4	(38.9, 90.6)	(41.5, 0.10)	(41.8, 180.1)	(38.8, 270.2)	-	-	-	-	-
5	(0.1, 211.1)	(50.7, 180.0)	(50.4, 359.9)	(50.6, 90.0)	(50.6, 270.0)	-	-	-	-
6	(17.2, 306.9)	(54.5, 266.1)	(22.5, 140.4)	(52.2, 360.0)	(52.4, 84.0)	(55.8, 185.2)	-	-	-
7	(5.4, 305.9)	(53.8, 170.7)	(27.8, 43.8)	(58.9, 355.7)	(54.4, 96.5)	(35.1, 220.0)	(54.8, 278.6)	-	-
8	(6.5, 318.3)	(61.5, 268.3)	(27.3, 317.0)	(51.8, 89.4)	(57.8, 5.80)	(53.7, 171.3)	(38.0, 200.3)	(39.3, 91.1)	-
9	(2.50, 28.2)	(36.5, 266.8)	(33.9, 182.3)	(56.9, 178.7)	(42.2, 78.4)	(53.1, 97.5)	(57.9, 359.7)	(37.1, 355.1)	(58.2, 272.1)

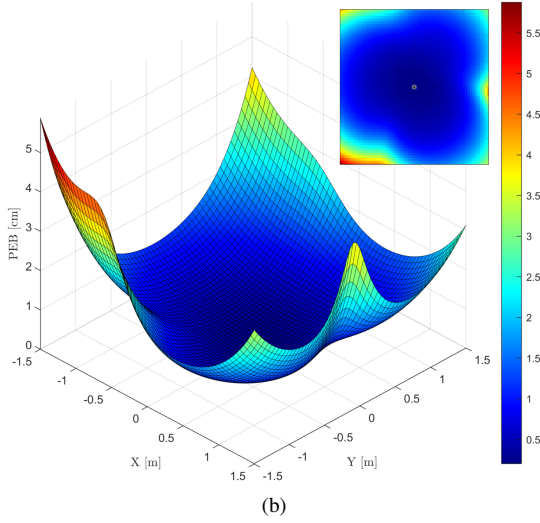
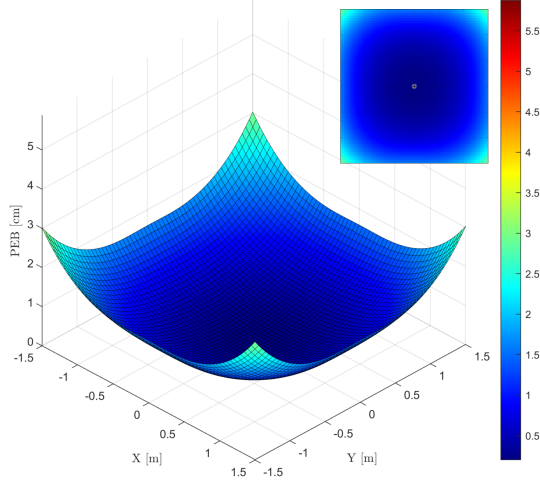


Fig. 3. Heat map of the PEB across a $3 \times 3 \text{ m}^2$ testbed at $z = 0.8 \text{ m}$ for $K = 5$ orientations: (a) optimal set, (b) random set.

for the optimal K -orientation set under three $\Phi_{1/2}$ scenarios at an effective SNR (from (6c)) of 14 dB, selected as a representative moderate-SNR setting and consistent with prior indoor VLP evaluations that analyze performance around 15 dB [46]. In all scenarios, $\text{PEB}_{90\%}$ decreases as K increases, indicating improved localization precision with additional orientations. Assuming a fixed per-orientation acquisition time, the total acquisition latency scales linearly with K ; under this accuracy–latency tradeoff, $K = 5$ offers a favorable compromise. For $\Phi_{1/2} = 30^\circ$, at least four orientations are

necessary due to the increased directivity limiting coverage with only three orientations; configurations with $K \geq 4$ outperform alternatives. In contrast, $\Phi_{1/2} = 45^\circ$ achieves competitive error performance starting at $K = 3$, and given its widespread use in commercial LEDs, it is adopted for the subsequent simulations.

Figure 5 illustrates the dependence of the $\text{PEB}_{90\%}$ on the SNR for various optimized K -orientation sets. As the SNR increases, $\text{PEB}_{90\%}$ decreases monotonically. Moreover, at any fixed SNR, increasing K yields a lower PEB, and the spacing between the curves diminishes at high SNR, indicating reduced incremental benefit from additional orientations under noise-limited conditions.

V. NONLINEAR ESTIMATOR

The optical power collected by the PD depends nonlinearly on its relative position to the LED, through the distance d , the irradiance angle ϕ_i , and the incidence angle ψ . Denoting the orientation vector of the receiver and the transmitter as $\mathbf{n}_r = [\alpha, \beta, \gamma]^T$ and $\mathbf{n}_{t,i} = [a_i, b_i, c_i]^T$, respectively, from (3), we have:

$$\cos \psi = \mathbf{n}_r \cdot (-\mathbf{n}_d) = -\frac{\alpha x + \beta y + \gamma(z - H)}{d}, \quad (22a)$$

$$\cos \phi_i = \mathbf{n}_{t,i} \cdot \mathbf{n}_d = \frac{a_i x + b_i y + c_i(z - H)}{d}, \quad (22b)$$

where H is the LED height. Substituting (3), (22), and $d = \sqrt{x^2 + y^2 + (z - H)^2}$ into the received-power model (4), yields:

$$P_{r,i} = -C \frac{[a_i x + b_i y + c_i(z - H)]^m}{[x^2 + y^2 + (z - H)^2]^{\frac{m+3}{2}}} \times [\alpha x + \beta y + \gamma(z - H)] + n_i, \quad (23)$$

where $n_i \sim \mathcal{N}(0, \sigma^2/N)$ and C is defined in (8).

Since our goal is to recover the unit-norm direction vector $\mathbf{n}_d$ rather than the absolute coordinates, the variables $(x, y, z - H)$ must lie on the unit sphere $\mathbb{S}^2$ (i.e., $x^2 + y^2 + (z - H)^2 = 1$). Defining:

$$Q_i(x, y, z) = a_i x + b_i y + c_i(z - H), \quad (24)$$

$$L(x, y, z) = \alpha x + \beta y + \gamma(z - H), \quad (25)$$

the model in (23) can be compactly written as:

$$P_{r,i} = -C Q_i^m(x, y, z) L(x, y, z) + n_i. \quad (26)$$

This scenario exemplifies a classical estimation problem in which the observations of $P_{r,i}$ for each orientation i are

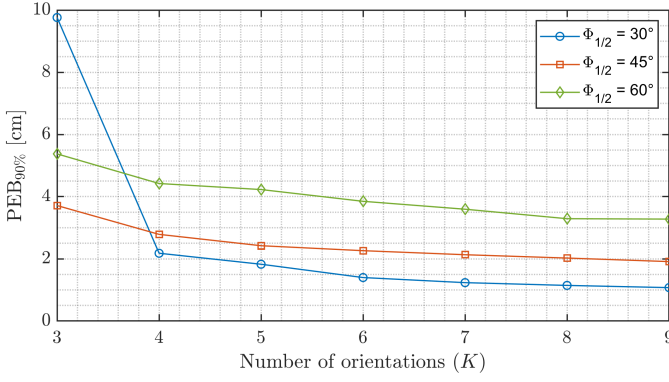


Fig. 4. $PEB_{90\%}$ versus the number of orientations K for GA-optimized sets at a fixed SNR of 14 dB and $\Phi_{1/2} \in \{30^\circ, 45^\circ, 60^\circ\}$. Values are computed over all receiver positions in the $3 \times 3 \times 2$ m³ testbed.

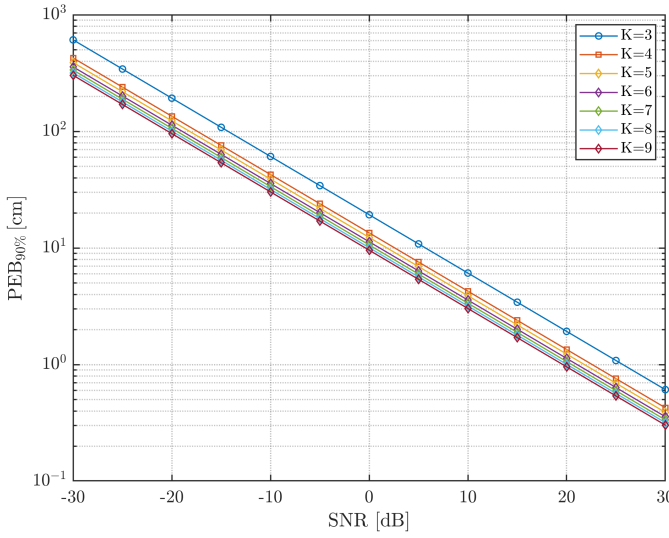


Fig. 5. $PEB_{90\%}$ versus SNR for GA-optimized orientation sets with $K \in \{3, \dots, 9\}$ ($\Phi_{1/2} = 45^\circ$; remaining parameters in Table II). Values are computed over all receiver positions in the $3 \times 3 \times 2$ m³ testbed.

nonlinearly related to the unknown receiver coordinates and corrupted by AWGN of known variance. Although no unbiased efficient estimator exists for this NL model, one can derive a MVU estimate by solving a constrained NL least-squares problem that minimizes the following cost function:

$$F(x, y, z) = \sum_{i=1}^K F_i(x, y, z), \quad (27)$$

where:

$$F_i(x, y, z) = \sum_{k=1}^N (P_{r,i,k} + C Q_i^m(x, y, z) L(x, y, z))^2. \quad (28)$$

Here $P_{r,i,k}$ denotes the k -th instantaneous received power sample at transmitter's orientation i . Moreover, the polynomial terms $Q_i(x, y, z)$ and $L(x, y, z)$ must satisfy $Q_i(x, y, z) \geq 0$ for $i = 1, \dots, K$ and $L(x, y, z) \leq 0$, corresponding respectively to $\cos \phi_i \geq 0$ and $\cos \psi \geq 0$ for incidence and irradiance angles in $[0^\circ, 90^\circ]$.

Thus, the NL direction-finding estimator is:

$$\hat{\mathbf{n}}_{d,NL} = \arg \min_{\substack{(x,y,z-H) \in \mathbb{S}^2 \\ Q_i(x,y,z) \geq 0, \quad i=1,\dots,K \\ L(x,y,z) \leq 0}} F(x, y, z), \quad (29)$$

where $\hat{\mathbf{n}}_{d,NL}$ denotes the estimated direction vector obtained with the NL method. In practice, this optimisation can be performed in MATLAB using the `solve` function, which internally invokes `fmincon` and employs gradient-descent algorithms suitable for constrained NL problems.

VI. LINEAR ESTIMATORS

The present section develops the direction-finding stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalized least squares, i.e., GLS, which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares, i.e., WLS, a less expensive approximation that neglects the weak cross-correlation between equations. For implementation details, an end-to-end pseudocode of the full procedure (GLS/WLS direction finding, distance recovery, and 3D position estimation) is provided in Algorithm 1.

A. Direction Estimation via GLS

This subsection consolidates the analytical procedure that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{n}_d$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

1) *Lambertian Power Ratios and Homogeneous Constraints:* Let $\mu_i = \mathbb{E}[P_{r,i}]$ be the noise-free average power at orientation i . We define power ratio β_i as:

$$\beta_i = \left(\frac{\mu_i}{\mu_1} \right)^{1/m}, \quad i = 2, \dots, K, \quad (30)$$

and, from (4), (2), (3) and (8):

$$\mu_i = -C \frac{(\mathbf{n}_{t,i} \cdot \mathbf{d})^m (\mathbf{n}_r \cdot \mathbf{d})}{d^{m+3}}. \quad (31)$$

Using (31) in (30), each β_i imposes the orthogonality condition:

$$\mathbf{a}_i \cdot \mathbf{d} = 0, \quad i = 2, \dots, K, \quad (32)$$

i.e., $\mathbf{a}_i \perp \mathbf{d}$, with:

$$\mathbf{a}_i = \mathbf{n}_{t,i} - \beta_i \mathbf{n}_{t,1}, \quad (33)$$

which therefore supplies $K - 1$ homogeneous hyperplanes that intersect at the true $\mathbf{d}$.

2) *Stochastic Perturbation of the Ratios:* Recall from (5) that the sample mean is:

$$\bar{P}_{r,i} = \frac{1}{N} \sum_{k=1}^N P_{r,k,i} \sim \mathcal{N}(\mu_i, \sigma^2/N). \quad (34)$$

For notational convenience, define $\hat{\mu}_i \triangleq \bar{P}_{r,i}$, which is the MVU estimator of μ_i . Injecting $\hat{\mu}_i$ into (30) yields:

$$\hat{\beta}_i = g(\hat{\mu}_1, \hat{\mu}_i), \quad g(x_1, x_2) = \left(\frac{x_2}{x_1} \right)^{\frac{1}{m}}. \quad (35)$$

As demonstrated in Appendix A, a first-order Taylor expansion of g centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\top$ gives:

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1} \right). \quad (36)$$

Hence every $\hat{\beta}_i$ shares the common perturbation n_1 , producing correlation across orientations.

Let us now define the ratio error:

$$\tilde{n}_i = \hat{\beta}_i - \beta_i. \quad (37)$$

Then, as demonstrated in Appendix B, using (36) and the independence of n_i and n_1 leads to:

$$\text{Var}[\tilde{n}_i] = \frac{\sigma^2}{Nm^2} \beta_i^2 (\mu_i^{-2} + \mu_1^{-2}), \quad (38)$$

$$\text{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{Nm^2} \beta_i \beta_j \mu_1^{-2}, \quad i \neq j. \quad (39)$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (38) demonstrates that the variance of $\tilde{n}_i$ depends explicitly on μ_i , even though each n_i has the same variance σ^2 . In other words, a single-LED system exhibits heteroscedastic ratio errors.

3) *Residuals and Their Covariance*: Define the empirical constraint normals as $\hat{\mathbf{a}}_i = \mathbf{n}_{t,i} - \hat{\beta}_i \mathbf{n}_{t,1}$ and the corresponding residuals as:

$$\xi_i = \hat{\mathbf{a}}_i \cdot \mathbf{d}, \quad i = 2, \dots, K. \quad (40)$$

Using (37) and (33), we have $\hat{\mathbf{a}}_i = \mathbf{a}_i - \tilde{n}_i \mathbf{n}_{t,1}$. Since $\mathbf{a}_i \cdot \mathbf{d} = 0$ by (32), it follows that:

$$\xi_i = -(\mathbf{n}_{t,1} \cdot \mathbf{d}) \tilde{n}_i = -\kappa \tilde{n}_i, \quad \kappa = \mathbf{n}_{t,1} \cdot \mathbf{d}. \quad (41)$$

Stacking $\boldsymbol{\xi} = [\xi_2, \dots, \xi_K]^\top$ and $\tilde{\mathbf{n}} = [\tilde{n}_2, \dots, \tilde{n}_K]^\top$ yields:

$$\begin{aligned} \boldsymbol{\xi} &= -\kappa \tilde{\mathbf{n}}, \\ \mathbb{E}[\boldsymbol{\xi}] &= \mathbf{0}_{1 \times (K-1)}, \\ \boldsymbol{\Sigma}_{\boldsymbol{\xi}} &= \text{Cov}(\boldsymbol{\xi}) = \kappa^2 \boldsymbol{\Sigma}_{\beta}, \end{aligned} \quad (42)$$

where $\boldsymbol{\Sigma}_{\beta} \in \mathbb{R}^{(K-1) \times (K-1)}$ collects the variances and covariances of $\tilde{n}_i$ given in (38)–(39).

4) *GLS Formulation and Closed-Form Solution*: Stack the empirical constraint normals into $\hat{\mathbf{A}} = [\hat{\mathbf{a}}_2 \dots \hat{\mathbf{a}}_K] \in \mathbb{R}^{3 \times (K-1)}$. From (40)–(42), the stacked residuals satisfy $\boldsymbol{\xi} = \hat{\mathbf{A}}^\top \mathbf{d}$ with $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}_{1 \times (K-1)}, \boldsymbol{\Sigma}_{\boldsymbol{\xi}})$ and $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2 \boldsymbol{\Sigma}_{\beta}$, $\kappa = \mathbf{n}_{t,1} \cdot \mathbf{d}$. Given an observation of $\boldsymbol{\xi} = \hat{\mathbf{A}}^\top \mathbf{d}$, the maximum-likelihood (equivalently, GLS) estimate of the direction is obtained by minimizing the Mahalanobis norm of $\boldsymbol{\xi}$ [42]:

$$\begin{aligned} \hat{\mathbf{d}} &= \arg \min_{\|\mathbf{d}\|=1} \|\hat{\mathbf{A}}^\top \mathbf{d}\|_{\boldsymbol{\Sigma}_{\boldsymbol{\xi}}}^2 \\ &= \arg \min_{\|\mathbf{d}\|=1} \mathbf{d}^\top (\hat{\mathbf{A}} \boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1} \hat{\mathbf{A}}^\top) \mathbf{d}. \end{aligned} \quad (43)$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2 \boldsymbol{\Sigma}_{\beta}$, the unknown scale κ^2 cancels in (43); hence one may use $\boldsymbol{\Sigma}_{\beta}^{-1}$ in place of $\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}$ without affecting the minimizer. The problem is a constrained Rayleigh quotient whose solution is the unit eigenvector associated with the smallest eigenvalue of:

$$\mathbf{M}_{\text{GLS}} = \hat{\mathbf{A}} \boldsymbol{\Sigma}_{\beta}^{-1} \hat{\mathbf{A}}^\top \in \mathbb{R}^{3 \times 3}. \quad (44)$$

Algorithm 1 3D Position Estimation from K Orientations

Require: Power samples $P_{r,i,k}$ ($k = 1, \dots, N$, $i = 1, \dots, K$); steering vectors $\mathbf{n}_{t,i}$; Lambertian order m ; noise variance σ^2 ; $\text{MODE} \in \{\text{GLS}, \text{WLS}\}$; radiometric constant C .

Ensure: $\hat{\mathbf{r}}$ (3D position of the receiver)

```

1: Direction finding:
2:   Compute  $\hat{\mu}_i = \frac{1}{N} \sum_{k=1}^N P_{r,k,i}$  for  $i = 1, \dots, K$ 
3:   Compute  $\hat{\beta}_i = (\hat{\mu}_i / \hat{\mu}_1)^{1/m}$  for  $i = 2, \dots, K$ 
4:   Form  $\mathbf{a}_i = \mathbf{n}_{t,i} - \hat{\beta}_i \mathbf{n}_{t,1}$ 
5:   Set  $\mathbf{A} = [\mathbf{a}_2, \dots, \mathbf{a}_K]$ 
6:   if  $\text{MODE} = \text{GLS}$  then
7:     Build  $\boldsymbol{\Sigma}_{\beta}$  via (38)–(39), substituting  $\beta_i \leftarrow \hat{\beta}_i$ 
8:      $\mathbf{M} \leftarrow \mathbf{A} \boldsymbol{\Sigma}_{\beta}^{-1} \mathbf{A}^\top$ 
9:   else ▷ WLS (diagonal approximation)
10:     $\tilde{\boldsymbol{\Sigma}}_{\beta} \leftarrow \text{diag}\{\hat{\beta}_i^2 (\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2})\}$ 
11:     $\mathbf{M} \leftarrow \mathbf{A} \tilde{\boldsymbol{\Sigma}}_{\beta}^{-1} \mathbf{A}^\top$ 
12:   end if
13:   Compute  $\mathbf{v}_{\min}$  = eigenvector of  $\mathbf{M}$  with smallest eigenvalue
14:    $\hat{\mathbf{n}}_d \leftarrow \mathbf{v}_{\min} / \|\mathbf{v}_{\min}\|$ 
15:   if  $\hat{\mathbf{n}}_d \cdot \mathbf{n}_{t,1} < 0$  then
16:      $\hat{\mathbf{n}}_d \leftarrow -\hat{\mathbf{n}}_d$  ▷ Ensure pointing to Rx
17:   end if

18: Distance estimation (beamsteering):
19:   Set  $\mathbf{n}_t \leftarrow \hat{\mathbf{n}}_d$ ,  $\mathbf{n}_r \leftarrow -\hat{\mathbf{n}}_d$ 
20:   Collect  $N$  power samples  $P_{r,K+1}$ 
21:   Compute  $\hat{\mu}_{K+1} = \frac{1}{N} \sum_{k=1}^N P_{r,K+1,k}$ 
22:    $\hat{d} \leftarrow \sqrt{\frac{C}{\hat{\mu}_{K+1}}}$ 

23: Final 3D position:
24:    $\hat{\mathbf{r}} \leftarrow \mathbf{t} + \hat{d} \hat{\mathbf{n}}_d$ 
25: return  $\hat{\mathbf{r}}$ 

```

Therefore:

$$\hat{\mathbf{n}}_{d,\text{GLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})\|}, \quad \hat{\mathbf{n}}_{d,\text{GLS}} \cdot \mathbf{n}_{t,1} > 0, \quad (45)$$

where $\mathbf{v}_{\min}(\cdot)$ denotes the eigenvector corresponding to the smallest eigenvalue; the sign convention ensures the direction points from the transmitter toward the receiver.

B. WLS as a Practical Simplification

The GLS solution in (43) requires inverting the $(K-1) \times (K-1)$ covariance $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2 \boldsymbol{\Sigma}_{\beta}$, typically via a Cholesky factorization [47]. For real-time, low-power receivers, a lighter alternative is to neglect the off-diagonal terms in (39) and approximate:

$$\tilde{\boldsymbol{\Sigma}}_{\beta} = \text{diag}\{\beta_i^2 (\mu_i^{-2} + \mu_1^{-2})\}_{i=2}^K. \quad (46)$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2 \boldsymbol{\Sigma}_{\beta}$, the unknown scale κ^2 cancels, and we define the WLS weighting matrix as $\tilde{\boldsymbol{\Sigma}}_{\beta}^{-1}$.

Using the empirical normals $\hat{\mathbf{A}}$ from the previous subsection, the WLS criterion is the Rayleigh quotient:

$$J_{\text{WLS}}(\mathbf{d}) = \mathbf{d}^\top (\hat{\mathbf{A}} \tilde{\boldsymbol{\Sigma}}_{\beta}^{-1} \hat{\mathbf{A}}^\top) \mathbf{d}, \quad \|\mathbf{d}\| = 1, \quad (47)$$

and we introduce the associated WLS matrix:

$$\mathbf{M}_{\text{WLS}} = \hat{\mathbf{A}} \tilde{\boldsymbol{\Sigma}}_{\beta}^{-1} \hat{\mathbf{A}}^\top \in \mathbb{R}^{3 \times 3}. \quad (48)$$

The minimizer is the unit eigenvector corresponding to the smallest eigenvalue of $\mathbf{M}_{\text{WLS}}$:

$$\hat{\mathbf{n}}_{d,\text{WLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})\|}, \quad \hat{\mathbf{n}}_{d,\text{WLS}} \cdot \mathbf{n}_{t,1} > 0. \quad (49)$$

The neglected cross-covariances in (39) scale with μ_1^{-2} , whereas the retained diagonal terms scale with $\mu_i^{-2} + \mu_1^{-2}$. Thus, when the reference orientation $i = 1$ is chosen such that $\mu_1 \gg \mu_i$ for most i , the off-diagonal terms are negligible relative to the diagonal terms and WLS closely tracks GLS while avoiding a full covariance inversion. In more symmetric geometries where $\mu_i \approx \mu_1$ for many i , the cross-correlations are non-negligible and the full GLS should be used for optimal performance.

VII. ESTIMATORS COMPARISON

A. Positioning Performance

This section compares the model-based estimators introduced earlier (NL, WLS, and GLS). For each transmitter orientation i , the received optical power was simulated according to (4) and (2) using the system parameters in Table II, with a semi-angle at half power $\Phi_{1/2} = 45^\circ$. Zero-mean AWGN $n_i \sim \mathcal{N}(0, \sigma^2)$ was drawn i.i.d. across orientations and samples; the sample mean $\bar{P}_{r,i}$ was formed from N realizations as in (5). Position estimates were then computed over the 3D testbed specified in Table II, using the GA-optimized orientation sets in Table III. Estimates were compared against the ground-truth positions, and the 3D Euclidean position error was recorded for each trial. Figure 6 plots the cumulative distribution function (CDF) of the 3D position error for $K \in \{3, 5, 9\}$. The case $K = 3$ uses the deterministic baseline of [22], which was originally proposed for 2D estimation; however, by applying the singular value decomposition (SVD) to the matrix obtained by stacking the vectors in (33), we found it can be extended to 3D, but only for three orientations (i.e., without WLS/GLS weighting). The setting $K = 5$ is selected as a latency–accuracy compromise based on the CRLB analysis, and $K = 9$ represents the upper bound of the orientation counts studied in this paper. For $K = 3$, $K = 5$ (GLS, WLS), and $K = 9$ (GLS, WLS), the GA-optimized orientation sets from Table III were used.

The NL estimator exhibited degraded performance when paired with those K -optimal sets because it minimizes (29) without an explicit weighting matrix, thereby giving comparable influence to orientations with negligible received power (e.g., $\phi_i > 90^\circ$), which impairs convergence. To provide a fair comparison, a separate GA optimization was conducted using the parameters in Table II but with the NL method’s RMSE as the objective; this NL-specific optimal set is used in the NL curves reported.

Across all estimators, the $K = 9$ configurations (dashed traces) stochastically dominate their $K = 5$ counterparts (solid lines), in agreement with the CRLB trend that additional orientations reduce the PEB. Among the model-based schemes, GLS consistently tracks the theoretical PEB most closely, regardless of K , with WLS incurring a small but visible loss due to its diagonal covariance approximation. All proposed

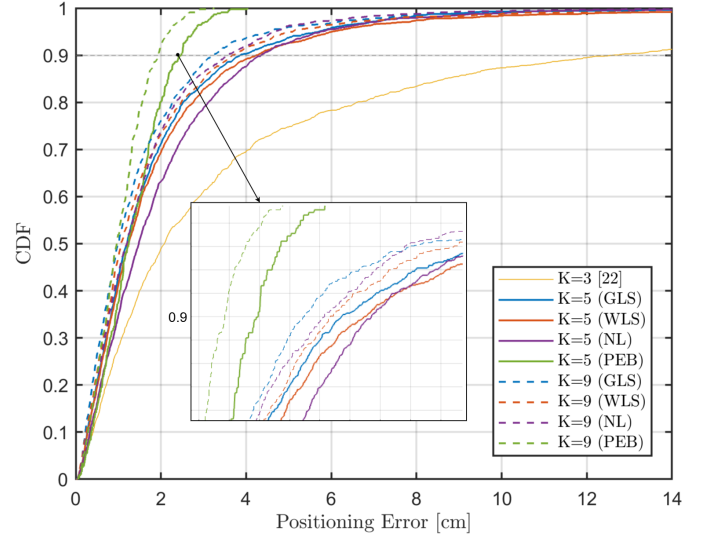


Fig. 6. Comparison of the CDFs of different 3D position estimation methods for $K = 3, 5, 9$.

estimators substantially outperform the deterministic $K = 3$ baseline [22].

Table IV summarizes the quantitative metrics: RMSE, the 90th percentile of the error CDF ($\text{CDF}_{90\%}$), and APE. For $K = 5$, GLS achieved $\text{RMSE} = 2.72$ cm and $\text{CDF}_{90\%} = 3.86$ cm, within a factor $\approx 1.75\times$ of the CRLB benchmark (1.55 cm RMSE). WLS trailed GLS by $\approx 36\%$ in RMSE at $K = 5$, reflecting the benefit of modeling inter-orientation error correlations. The NL estimator, after re-optimizing its orientation set, became competitive with WLS at $K = 5$ and surpassed WLS at $K = 9$, where NL attained an RMSE of 2.36 cm versus 2.96 cm for WLS. Increasing K from 5 to 9 delivered consistent gains for all methods (e.g., GLS RMSE improved from 2.72 cm to 2.41 cm), and the gap to the CRLB narrowed (from 1.55 cm to 1.21 cm). Overall, GLS provided the best fidelity to the bound with modest complexity; WLS offered a favorable accuracy–complexity compromise; and NL benefited markedly from orientation-set co-design, especially at larger K . However, NL does not surpass GLS, despite appearing more general, NL is unweighted and does not account for the relative contribution of each constrain equation in the position solve, and therefore it fails to outperform the weighted GLS method. The APE values track the same ordering as RMSE and $\text{CDF}_{90\%}$, indicating that the improvements are uniform across the distribution rather than confined to the median or the tail.

Figure 7 illustrates the 3D position estimates produced by GLS and WLS for $K = 5$ at three receiver heights $z \in \{0, 0.6, 1.2\}$ m, overlaid on the ground truth. At $z = 1.2$ m, errors increase near the testbed boundaries—particularly at the corners—consistent with the PEB heat maps, shown in Fig. 3(a), and attributable to reduced SNR at extreme incidence angles. At $z = 0$ m, the errors are more spatially uniform. In both cases, GLS maintains a tighter scatter than WLS, in line with its closer adherence to the CRLB.

TABLE IV
PERFORMANCE OF POSITION-ESTIMATION
METHODS IN A $3 \times 3 \times 2 \text{ m}^3$ ROOM.

K	Method	RMSE [cm]	CDF 90% [cm]	APE [cm]
3	[22]	11.05	12.61	5.35
5	GLS	2.72	3.86	1.79
5	WLS	3.71	4.24	2.02
5	NL	2.90	4.35	2.05
5	CRLB	1.55	2.48	1.34
9	GLS	2.41	3.23	1.54
9	WLS	2.96	3.67	1.75
9	NL	2.36	3.57	1.64
9	CRLB	1.21	1.93	1.04

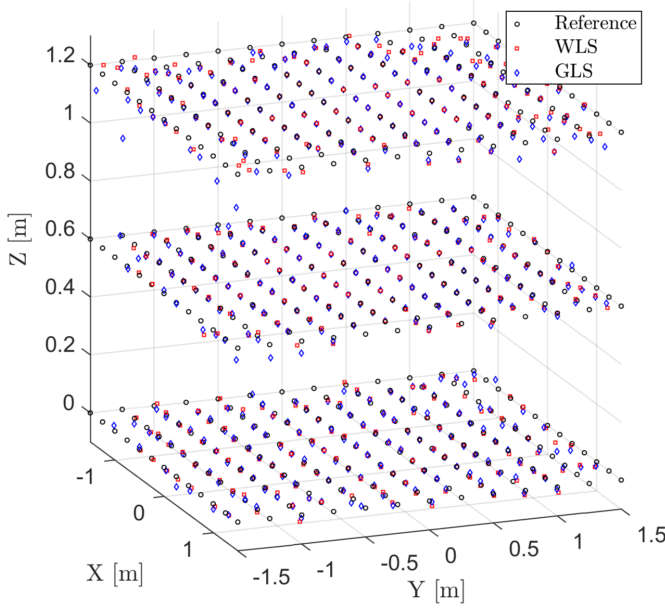


Fig. 7. 3D position estimates at three receiver heights for $K = 5$ orientations: GLS, WLS, ground truth.

B. Computational Complexity

We summarize the per-inference computational cost for direction finding under K LED orientations and N samples per orientation; computing the sample means $\{\hat{\mu}_i\}_{i=1}^K$ via (5) incurs $\mathcal{O}(NK)$ and is common to all schemes. We next detail the direction-finding estimators (GLS/WLS/NL); the subsequent distance-recovery step is identical across methods and requires constant time, as given by (9).

1) *GLS*: From (38)–(39) we assemble the full covariance $\Sigma_\beta \in \mathbb{R}^{(K-1) \times (K-1)}$ (cost $\mathcal{O}(K^2)$), then factor it (Cholesky) and apply the solves needed to build $\mathbf{M}_{\text{GLS}} = \hat{\mathbf{A}} \Sigma_\beta^{-1} \hat{\mathbf{A}}^\top$ in (44). The factorization dominates with $\mathcal{O}((K-1)^3)$ flops; the subsequent triangular solves are $\mathcal{O}(K^2)$. The eigenproblem is 3×3 and thus constant-cost. Overall, the cost per position estimate is $\mathcal{O}(NK) + \mathcal{O}(K^2) + \mathcal{O}(K^3)$ with GLS.

2) *WLS*: Replacing Σ_β by the diagonal $\tilde{\Sigma}_\beta$ in (46) removes the full covariance inversion. Forming $\mathbf{M}_{\text{WLS}} = \hat{\mathbf{A}} \tilde{\Sigma}_\beta^{-1} \hat{\mathbf{A}}^\top$ requires only column scalings and a $3 \times (K-1)$ -by- $(K-1) \times 3$ product, i.e., $\mathcal{O}(K)$. The final eigenproblem is again constant-cost. Hence, the cost per position estimate is $\mathcal{O}(NK) + \mathcal{O}(K)$ with WLS.

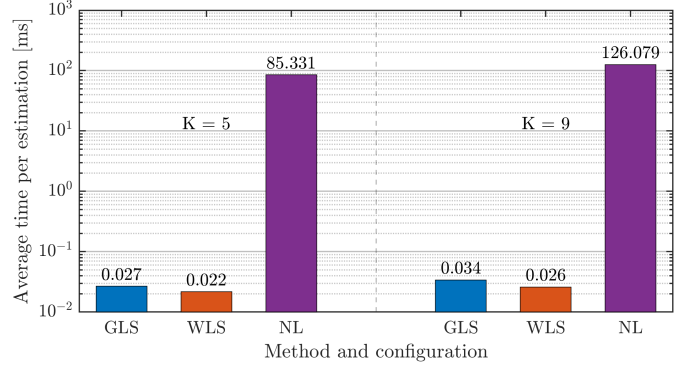


Fig. 8. Computation latency of the NL, GLS, and WLS estimators for $K = 5$ and $K = 9$.

3) *NL*: Solving (29) with MATLAB's `fmincon` (interior-point) entails, at each iteration, evaluating residuals and Jacobians for K orientations and solving a constant-size normal equation; the latter is constant-cost, while Jacobian/residual assembly is $\mathcal{O}(K)$. Using the Gauss–Newton least-squares structure for cost accounting, and with T_{NL} interior-point iterations, the per-estimate cost is $\mathcal{O}(NK) + \mathcal{O}(T_{\text{NL}}K)$.

Figure 8 reports the average end-to-end computation time per 3D estimate (powers pre-acquired). For $K=5$, WLS and GLS complete in 0.022 ms and 0.027 ms, respectively, while NL requires 85.331 ms; for $K=9$, WLS and GLS remain in the tens of microseconds (0.026 ms and 0.034 ms), whereas NL increases to 126.079 ms. The growth from $K=5$ to $K=9$ is modest for WLS (+18%) and GLS (+26%), consistent with their $\mathcal{O}(K)$ and small $\mathcal{O}(K^3)$ components, noting that the cubic term in GLS acts on a compact $(K-1) \times (K-1)$ system followed by a constant-cost 3×3 eigensolve. By contrast, NL is two to three orders of magnitude slower and increases by $\approx 48\%$ with K due to the constrained nonlinear solve performed with MATLAB's `fmincon` (interior-point), which repeatedly evaluates K -sized Jacobians as part of the iteration. These measurements corroborate the complexity analysis above: closed-form WLS/GLS enable real-time inference without fingerprinting or model retraining, whereas NL is latency-limited.

VIII. CONCLUSION AND FUTURE WORKS

This paper introduced a beam-steered single-LED, single-PD OWP architecture that delivers full 3D indoor localization without receiver rotation, cameras, or PD arrays. Our method is explicitly model-based, grounded in radiometry and geometry, so it requires no fingerprinting or data retraining, enabling interpretable performance, principled uncertainty analysis, and efficient co-design of hardware and algorithms.

We proposed a two-stage process: (i) direction finding from K steered orientations, followed by (ii) distance recovery from a single beam-aligned power measurement. A full CRLB/PEB analysis characterized identifiability ($K \geq 3$) and showed the benefits of overdetermination ($K \geq 4$). A GA-based orientation-set optimizer minimized the RMSE PEB across a 3D testbed and yielded practical steering patterns. Additionally, we introduce a ratio-based linearization of the

Lambertian power model that yields closed-form GLS/WLS direction estimators for stage (i).

Simulations demonstrated centimeter-level accuracy with modest K . Optimized orientation sets achieved sub-1.5 cm median PEB across $K \in [3, 9]$, reaching 0.74 cm at $K = 9$. With $K = 5$, GLS attained RMSE 2.72 cm and $\text{CDF}_{90\%} = 3.86$ cm, tracking the CRLB within a factor $\approx 1.75\times$; increasing to $K = 9$ improved GLS to RMSE 2.41 cm. Across configurations, the proposed estimators substantially outperformed the deterministic $K = 3$ baseline [22]. Importantly, GLS/WLS reduce inference to a single 3×3 eigenproblem, yielding μs -level latency, while the NL estimator exhibits ms-level runtimes. These results confirm that closed-form, model-based estimation can closely approach the theoretical bound at real-time speeds.

Beyond accuracy, the architecture is deployment-friendly. By concentrating complexity in the infrastructure (beam steering at a single NIR source) and using only one PD per receiver, the system avoids dense LED constellations and specialized user hardware. In operation, the transmitter iteratively cycles through the predefined set of K steering orientations; this periodic scan can be exploited by multiple receivers present in the scene, enabling concurrent localization for pedestrians and other dense multi-receiver scenarios (e.g., robot swarms) with minimal on-robot implementation.

Possible future work includes augmenting the radiometric-geometric model with physics-informed neural networks to learn small residual corrections while preserving physical constraints, enabling a gray-box (rather than black-box) estimator. Future work also includes validating the approach experimentally and extending the framework to OWC, with emphasis on single-source optical integrated sensing and communication.

APPENDIX A

PROOF OF EQUATION (36)

For each orientation i , the sample mean $\hat{\mu}_i \triangleq \bar{P}_{r,i}$ satisfies $\hat{\mu}_i = \mu_i + n_i$ with $n_i \sim \mathcal{N}(0, \sigma^2/N)$ mutually independent [43]. For $i \geq 2$, define $g(x_1, x_2) = (x_2/x_1)^{1/m}$ so that $\hat{\beta}_i = g(\hat{\mu}_1, \hat{\mu}_i)$ and $\beta_i = g(\mu_1, \mu_i)$. A first-order multivariate Taylor expansion (delta method) of g at (μ_1, μ_i) gives:

$$\hat{\beta}_i \approx \beta_i + \frac{\partial g}{\partial x_1}(\mu_1, \mu_i) n_1 + \frac{\partial g}{\partial x_2}(\mu_1, \mu_i) n_i. \quad (50)$$

The partial derivatives of g are, for general (x_1, x_2) :

$$\frac{\partial g}{\partial x_1}(x_1, x_2) = -\frac{1}{m} x_2^{\frac{1}{m}} x_1^{-\frac{1}{m}-1} = -\frac{g(x_1, x_2)}{m x_1}, \quad (51a)$$

$$\frac{\partial g}{\partial x_2}(x_1, x_2) = \frac{1}{m} x_2^{\frac{1}{m}-1} x_1^{-\frac{1}{m}} = \frac{g(x_1, x_2)}{m x_2}. \quad (51b)$$

Evaluating at (μ_1, μ_i) (so $g(\mu_1, \mu_i) = \beta_i$) gives:

$$\frac{\partial g}{\partial x_1}(\mu_1, \mu_i) = -\frac{\beta_i}{m \mu_1}, \quad (52a)$$

$$\frac{\partial g}{\partial x_2}(\mu_1, \mu_i) = \frac{\beta_i}{m \mu_i}. \quad (52b)$$

Hence:

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1} \right), \quad (53)$$

which is (36). $\square$

APPENDIX B

PROOF OF EQUATION (39)

From the first-order expansion (delta method):

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1} \right), \quad n_\ell \sim \mathcal{N}\left(0, \frac{\sigma^2}{N}\right) \text{ i.i.d.}, \quad (54)$$

we can deduce that the ratio error is:

$$\tilde{n}_i \triangleq \hat{\beta}_i - \beta_i \approx \frac{\beta_i}{m} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1} \right). \quad (55)$$

Using $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$ for independent X, Y :

$$\text{Var}[\tilde{n}_i] = \left(\frac{\beta_i}{m} \right)^2 \left(\frac{\sigma^2}{N \mu_i^2} + \frac{\sigma^2}{N \mu_1^2} \right) = \frac{\sigma^2}{N m^2} \beta_i^2 (\mu_i^{-2} + \mu_1^{-2}), \quad (56)$$

which is (38). For $i \neq j$:

$$\begin{aligned} \text{Cov}[\tilde{n}_i, \tilde{n}_j] &= \left(\frac{\beta_i}{m} \right) \left(\frac{\beta_j}{m} \right) \text{Cov} \left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}, \frac{n_j}{\mu_j} - \frac{n_1}{\mu_1} \right) \\ &= \left(\frac{\beta_i \beta_j}{m^2} \right) \left[0 - 0 - 0 + \text{Var} \left(\frac{n_1}{\mu_1} \right) \right] \\ &= \frac{\sigma^2}{N m^2} \beta_i \beta_j \mu_1^{-2}, \end{aligned} \quad (57)$$

since n_i, n_j, n_1 are mutually independent and $\text{Var}(n_1/\mu_1) = \sigma^2/(N \mu_1^2)$. Thus:

$$\text{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{N m^2} \beta_i \beta_j \mu_1^{-2}, \quad i \neq j, \quad (58)$$

which is (39). $\square$

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